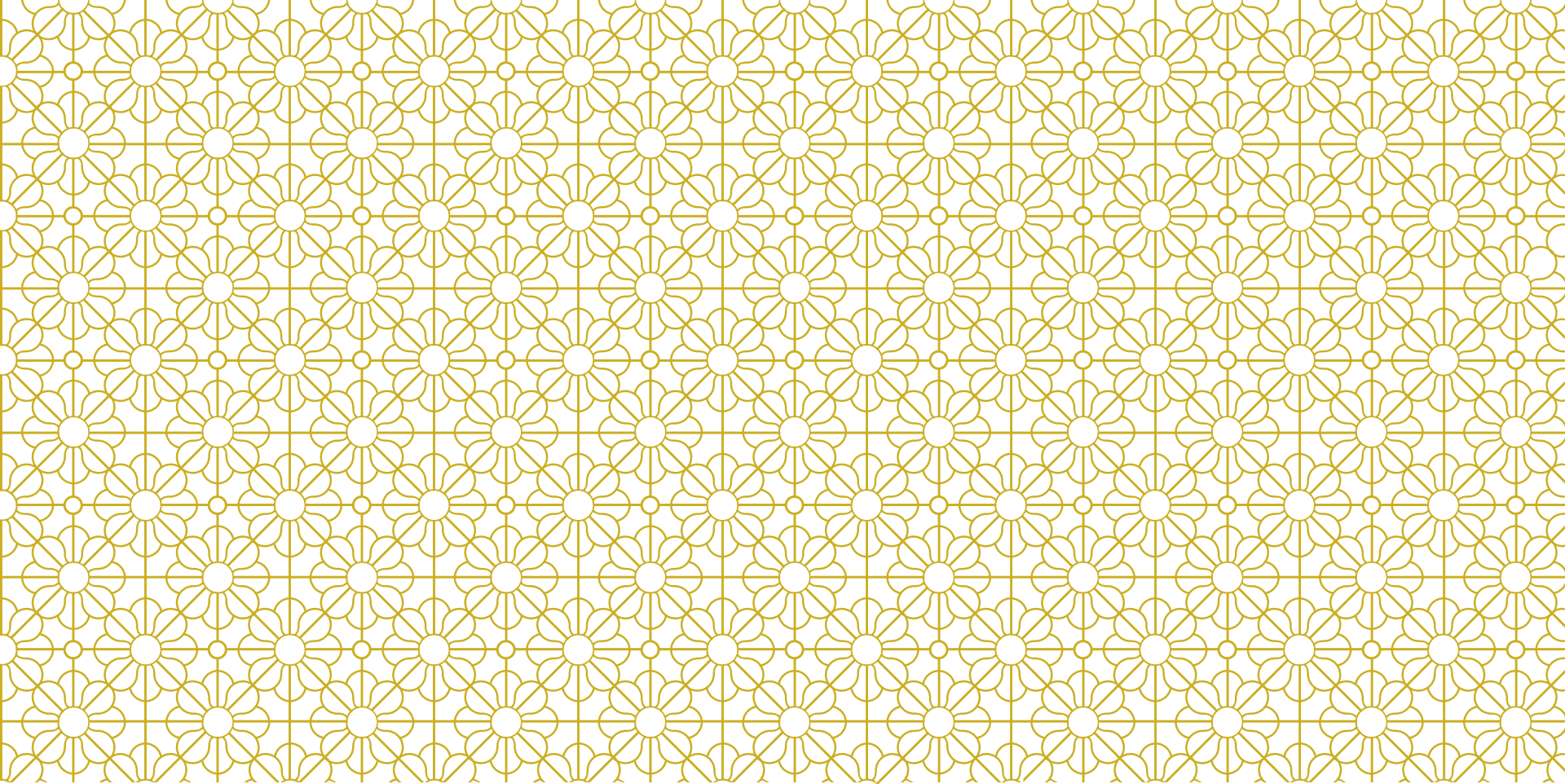




INVESTING LAYERS OF ENAMEL

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BDS, MDS

OBJECTIVES

By the end of the lecture student will be able to understand:

What are investing layers

Layers associated with the:

Unerupted tooth (Reduced enamel epithelium)

Erupting tooth crown (primary enamel cuticle)

Erupted crown(acquired pellicle, Dental Plaque , Calculus)

Clinical correlations

INTRODUCTION

Throughout its life, the crown of a tooth is covered by an organic layer or integument.

Before the tooth erupts into the oral cavity the crown is covered by the overlying *oral mucosa, the coronal part of the dental follicle, and the vestiges of the enamel organ* (plus its associated primary enamel cuticle).

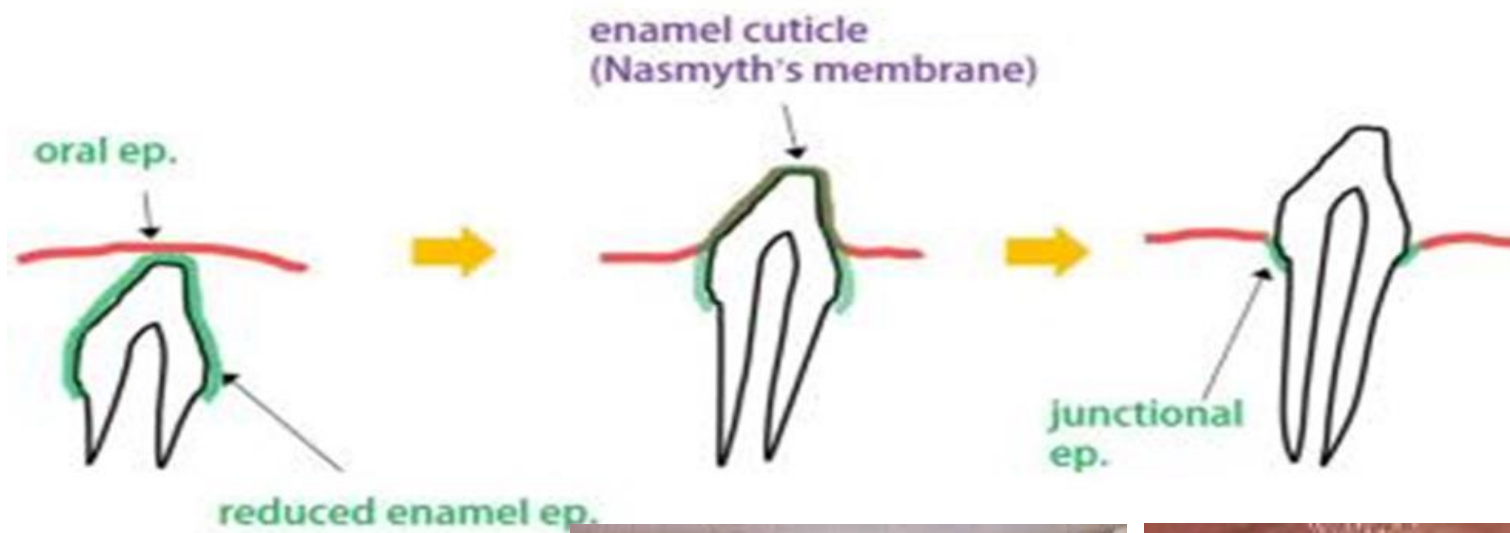
After emerging into the mouth, parts of the integument of enamel organ origin are lost by degeneration of its epithelial component and by attrition or abrasion of the underlying cuticular component.

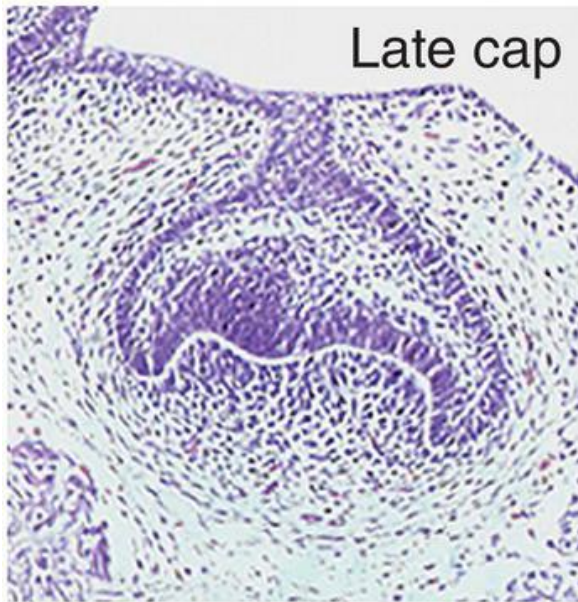
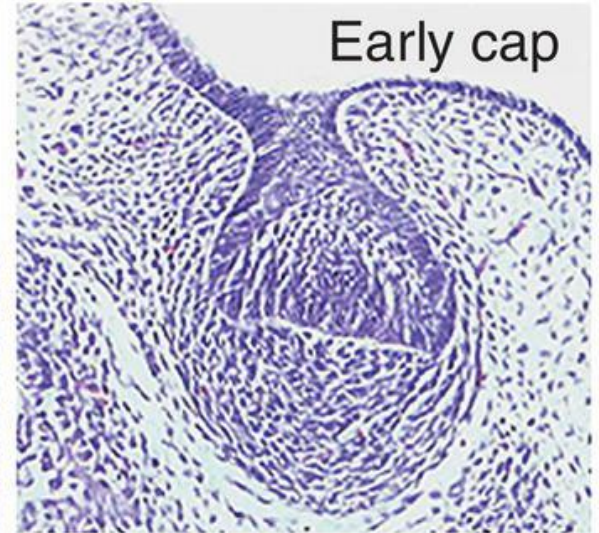
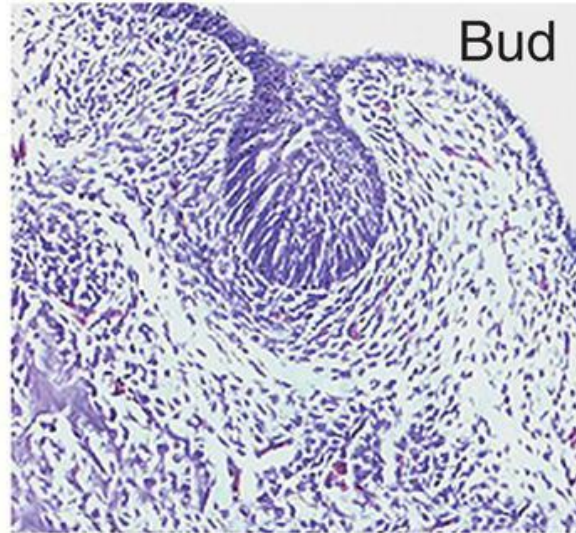
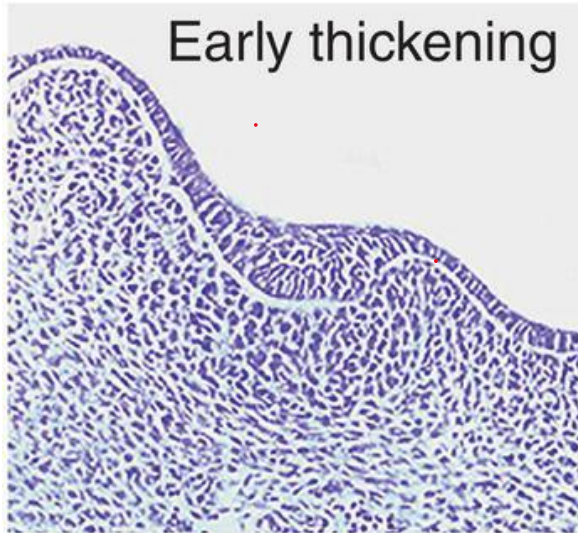
In the region of the gingival crevice or sulcus, the primary (or pre-eruptive) enamel cuticle acquires additional matter from the lining epithelium and, coronal to the gingival margin, from saliva.

The salivary layer is known as the *acquired pellicle*.

Oral bacteria adhere initially to the enamel cuticle and later to the acquired pellicle, to form the *dental plaque*.

Dental plaque may mineralize to form *dental calculus*.

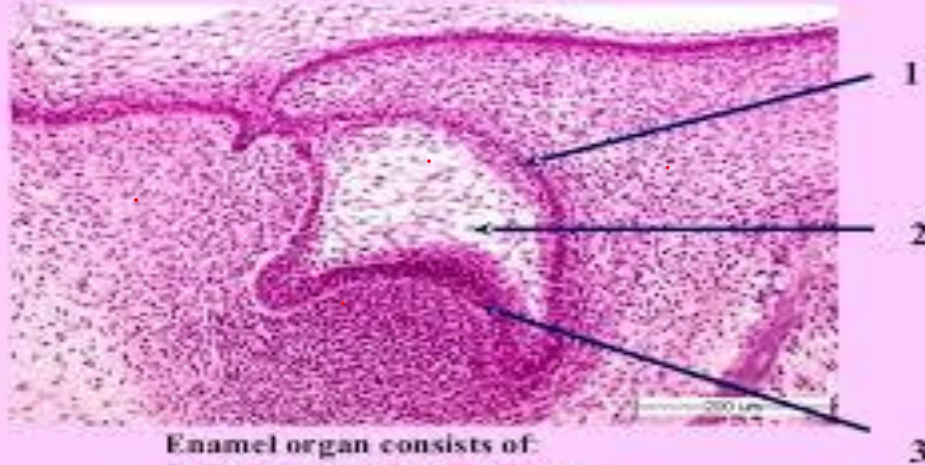




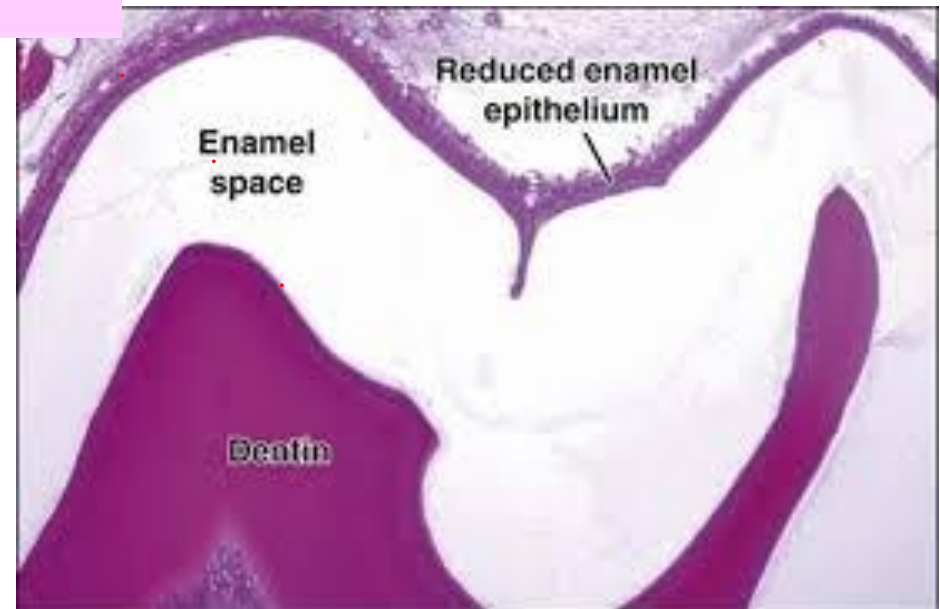
- oral epithelium
- dental lamina
- outer enamel epithelium
- stellate reticulum
- inner enamel epithelium
- dental follicle
- dental papilla
- cervical loop

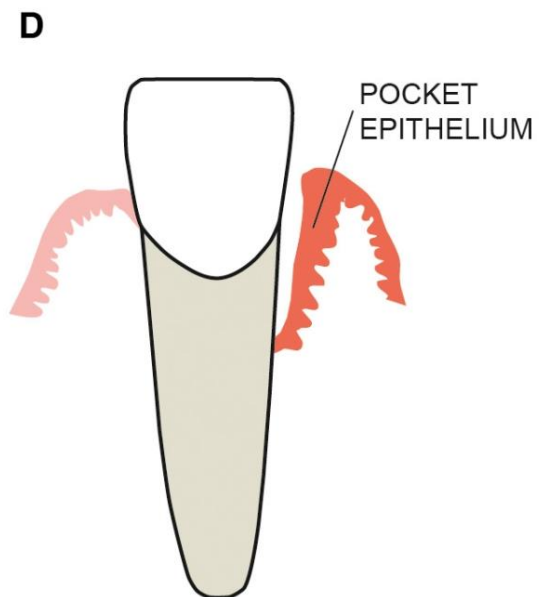
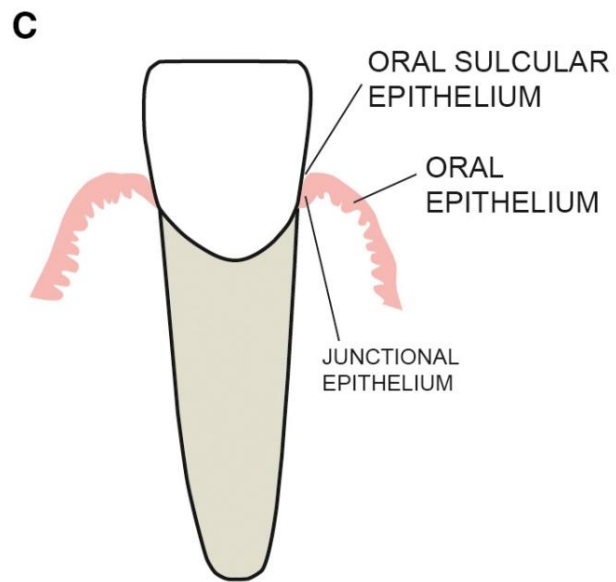
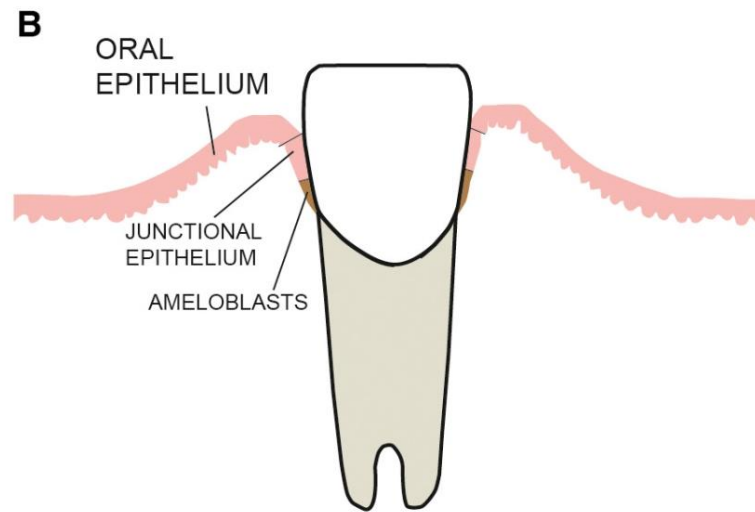
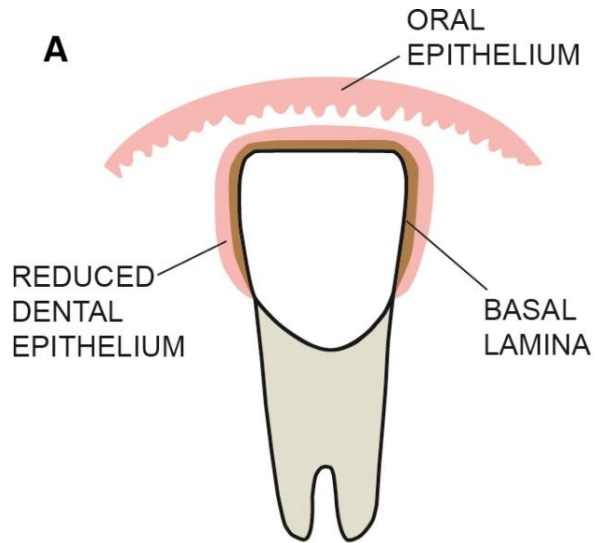
SOFT TISSUES COVERING AN UNERUPTED TOOTH

ENAMEL ORGAN

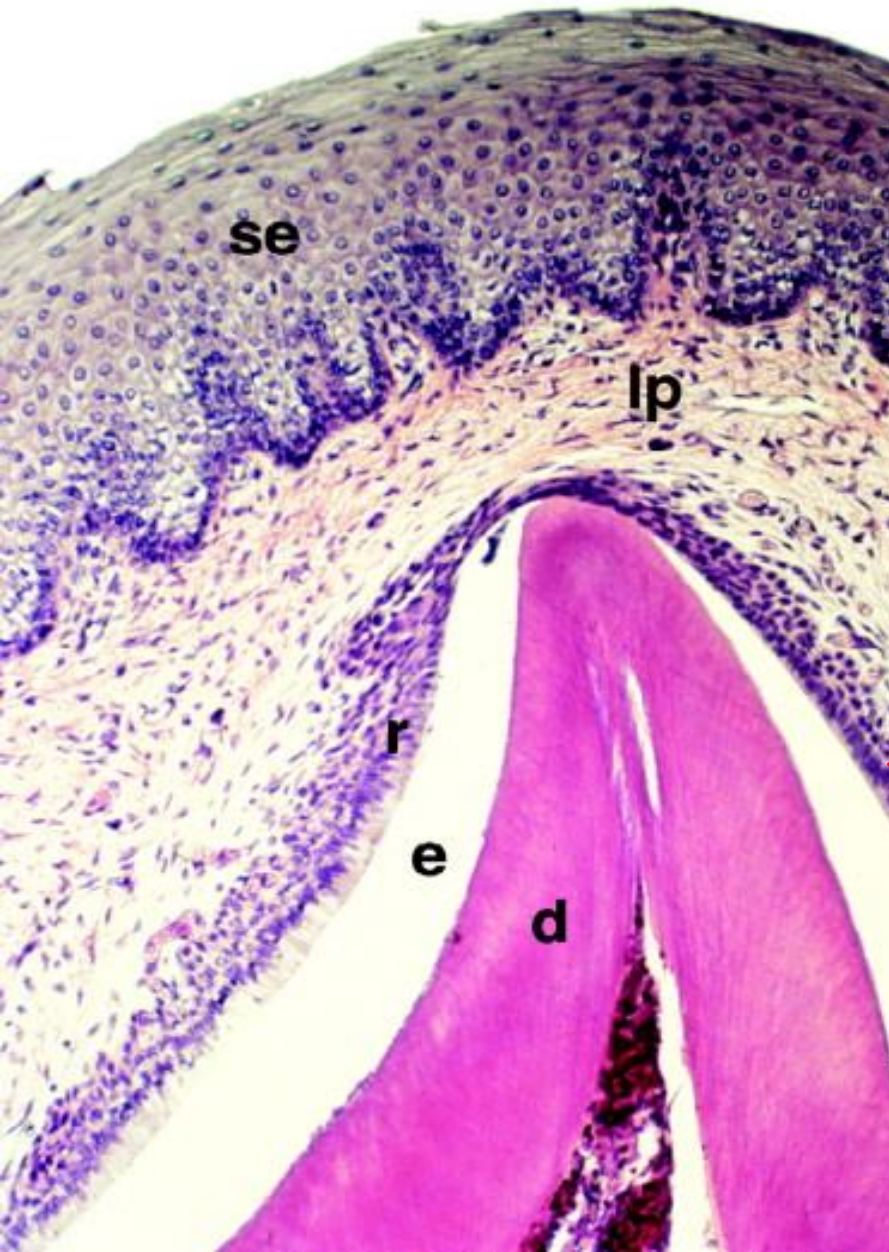


Enamel organ consists of:
1. Outer enamel epithelium
2. Stellate reticulum
3. Inner enamel epithelium





SOFT TISSUES COVERING AN UNERUPTED TOOTH



The soft tissues covering an unerupted tooth comprise **oral mucosa** and the subjacent connective tissue of the dental follicle.

Between the **dental follicle** and the enamel is an epithelial layer that is the remains of the **enamel organ** — the reduced enamel epithelium; a basal lamina (primary enamel cuticle) is interposed between the enamel surface and the reduced enamel epithelium.

Medium-power view showing the reduced enamel epithelium covering a tooth just before eruption.

d, dentin;

e, enamel space (enamel is lost during decalcification);

lp, lamina propria;

r, reduced enamel epithelium;

se, surface epithelium.

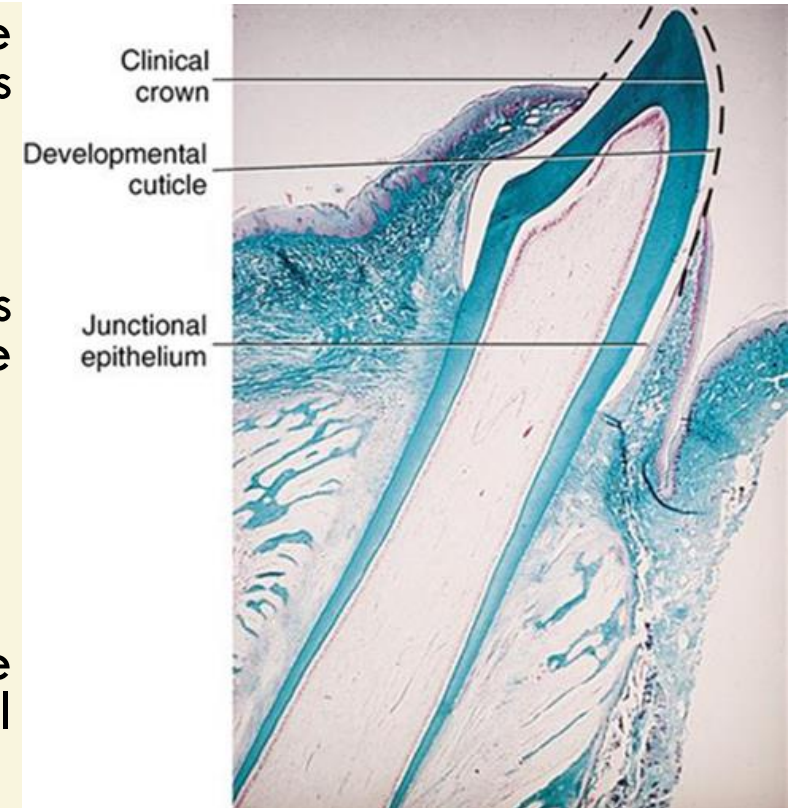
PRIMARY ENAMEL CUTICLE

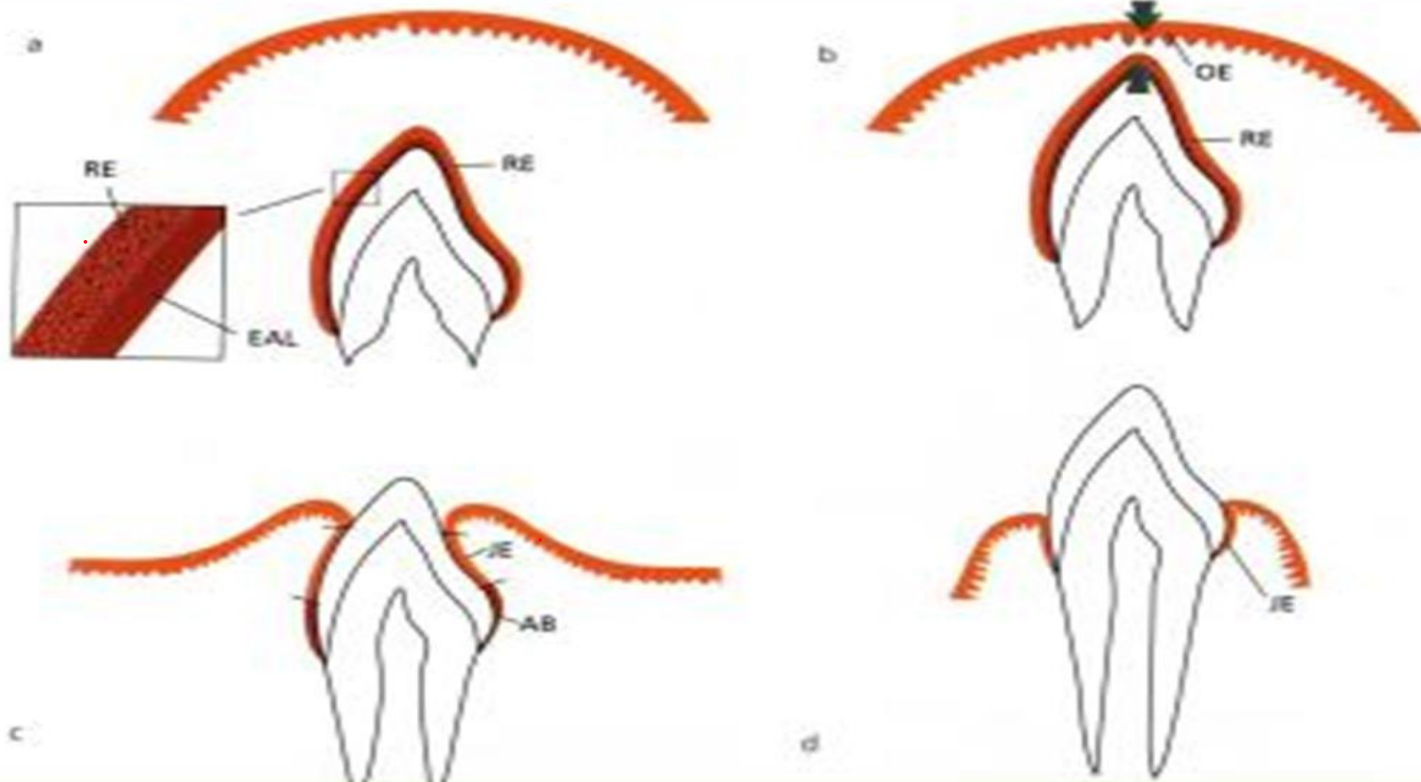
(Enamel cuticle/Nasmyth's membrane)

The reduced enamel epithelium and the primary enamel cuticle comprise Nasmyth's membrane.

The original reduced enamel epithelium is lost, leaving the primary enamel cuticle initially covering the exposed enamel.

The primary enamel cuticle is in intimate contact with the underlying organic enamel matrix.





- Covers the entire crown of the newly erupted tooth
- removed by mastication/attrition or abrasion
- Basal lamin secreted by the ameloblast after enamel formation completion
- Protects enamel surface from resorption by the vascular tissues prior to the eruption of tooth
- It is composed of protein mainly proteoglycan and glycoprotein elements from the contiguous soft tissues.
- Plasma contributions include immunoglobulins , which form part of the host defence system against plaque.

SOFT TISSUES COVERING AN ERUPTED TOOTH

A number of organic layers cover the erupted healthy tooth

- Pellicle
- Plaque
- Calculus

ACQUIRED PELLICLE

Where the enamel surface is exposed to wear, either by attrition or abrasion, the vestigial enamel organ is worn away, but the enamel rapidly acquires a layer of acquired pellicle.

Indeed, this pellicle always forms a protective coat following any wear.

This acellular layer is derived mainly from salivary proteins, but includes elements from crevicular fluid and bacteria.

This cuticle immediately acquires an organic element of salivary origin, the acquired pellicle.

DENTAL PLAQUE

Dental plaque is the combination of bacteria embedded in a matrix of salivary proteins and bacterial products superimposed on the acquired pellicle.

Dental plaque is an example of a biofilm, a term used to describe *communities of microbes attached to surfaces*.

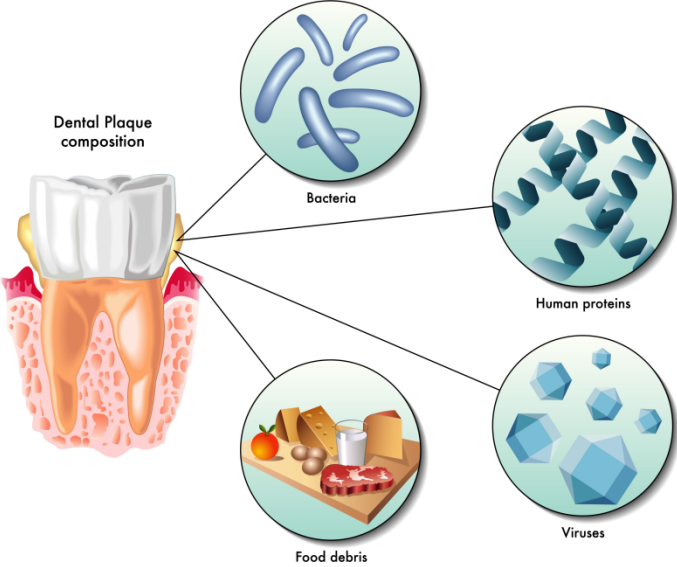
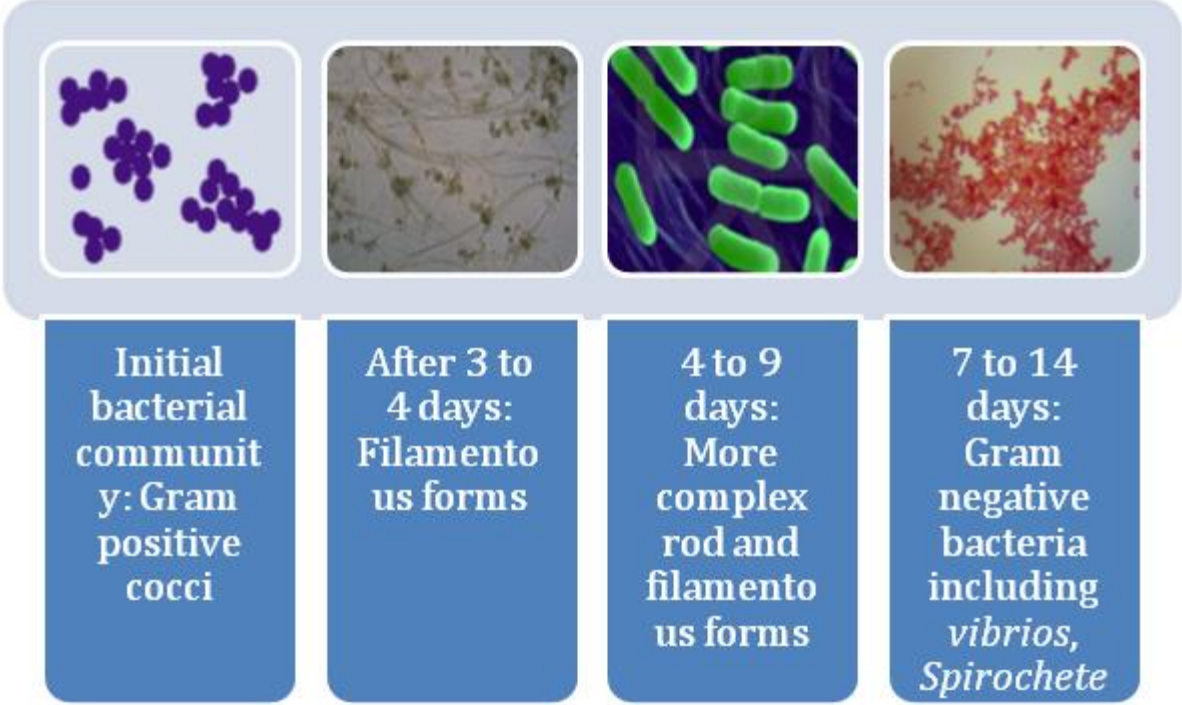
Early plaque is composed of mainly Gram-positive, facultative, anaerobic cocci and filaments.

With time, the deposit will thicken, although in non-pathological, supragingival situations its microfloral composition is unlikely to vary greatly.

- ❑ Plaque can be described as a soft, adherent, predominantly microbial mass which accumulates on the tooth surface in the absence of oral hygiene measures.
- ❑ Although it contains a large variety of micro-organisms, it is also composed of a rich organic matrix, derived from saliva, foodstuffs and microbial metabolism.



THE MICROBIAL COMPOSITION OF DENTAL PLAQUE WILL VARY, NOT ONLY WITH THE STAGES OF MATURITY OF THE DEPOSIT BUT ALSO FROM INDIVIDUAL TO INDIVIDUAL, FROM TOOTH TO TOOTH, FROM SITE TO SITE AND FROM SURFACE TO SURFACE.



with anaerobic Gram-negative organisms predominating, to include cocci, rods, filaments and many motile forms (particularly spirochaetes) .

Where the plaque is associated with chronic inflammatory periodontal disease and becomes subgingival, a more complex flora develops



- **Plaque can be seen on all tooth surfaces that are not subject to constant abrasion, especially in areas that are difficult to clean (such as occlusal pits and fissures, interproximal regions and at the gingival margin).**
- **Its presence may be more readily visualized clinically by the use of disclosing solutions (Fig. 8.19). Many plaque bacteria metabolize dietary carbohydrates, often producing polysaccharides, which may be stored intracellularly, on the cell surface and extracellularly in the matrix.**

BEFORE PLAQUE DISCLOSING TABLETS



AFTER PLAQUE DISCLOSING TABLETS

MECHANISMS OF PLAQUE FORMATION

It is clear that there are distinct stages to the formation of plaque:

Firstly, there is the initial transport of bacteria to tooth surface.

This is followed by reversible adsorption of the bacteria on to the pellicle surface.

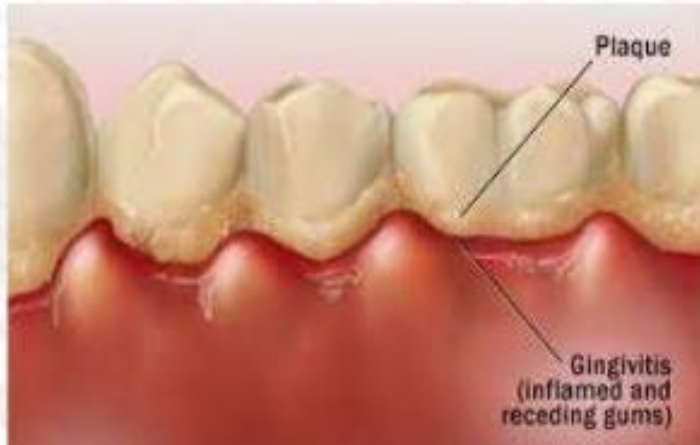
Less reversible attachment of bacteria to tooth surface then occurs.

These stages are repeated, forming new layers of bacterial adherence.

Finally, there is growth of the attached organisms, leading to the formation of a 'climax' community.

Steps of Plaque Formation	Description
Association	Dental pellicle forms on the tooth (normally on tooth), and provides bacteria surface to attach
Adhesion	Within hours, bacteria loosely binds to the pellicle.
Proliferation	Bacteria spreads throughout the mouth and begins to multiply.
Microcolonies	Microcolonies are formed. Streptococci secrete protective layer (slime layer).
Biofilm formation	Microcolonies form complex groups with metabolic advantages.
Growth or maturation	The biofilm develops a primitive circulatory system

SUPRAGINGIVAL PLAQUE



Tooth-Attached Plaque

Mostly gram+ bacteria; with some gram- cocci and rods.

Tooth attached plaque

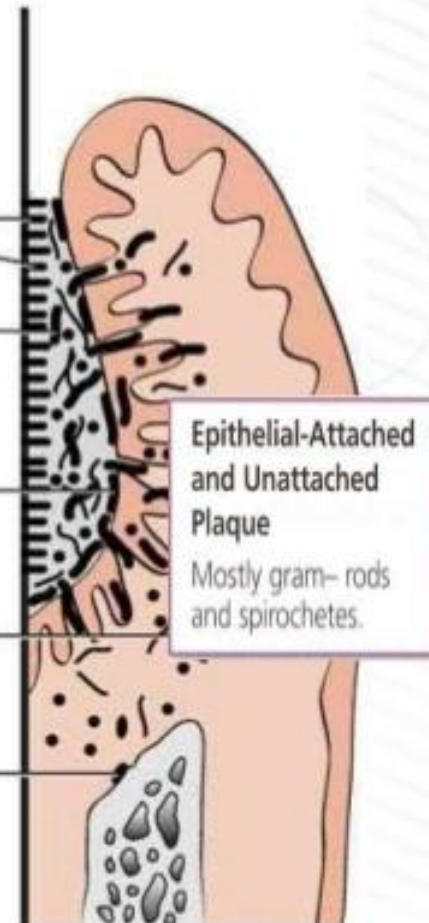
Unattached plaque

Epithelial associated plaque

Bacteria within connective tissue

Bacteria on bone surface

SUBGINGIVAL PLAQUE



DENTAL CALCULUS

Dental calculus is mineralized plaque and may become attached to the enamel of the crown or the cementum (or dentine) of the root.



- **Saliva is supersaturated with minerals (such as calcium and phosphate) that have the potential to mature newly erupted enamel, protect exposed tooth surfaces from acid action and remineralize areas in the early stages of demineralization.**
- **Salivary inhibitors prevent precipitation and crystallization of minerals in saliva, but bacterial enzymes can degrade these inhibitors.**
- **Thus, under suitable conditions (e.g. with high concentrations of minerals derived from saliva), precipitation and crystallization may occur within dental plaque.**

The organic component of dental calculus, being derived from plaque, will therefore be derived from :

saliva,

gingival crevicular fluid,

desquamated epithelial cells,

blood cells,

food debris,

bacteria and their products.

CLINICAL CONSIDERATION

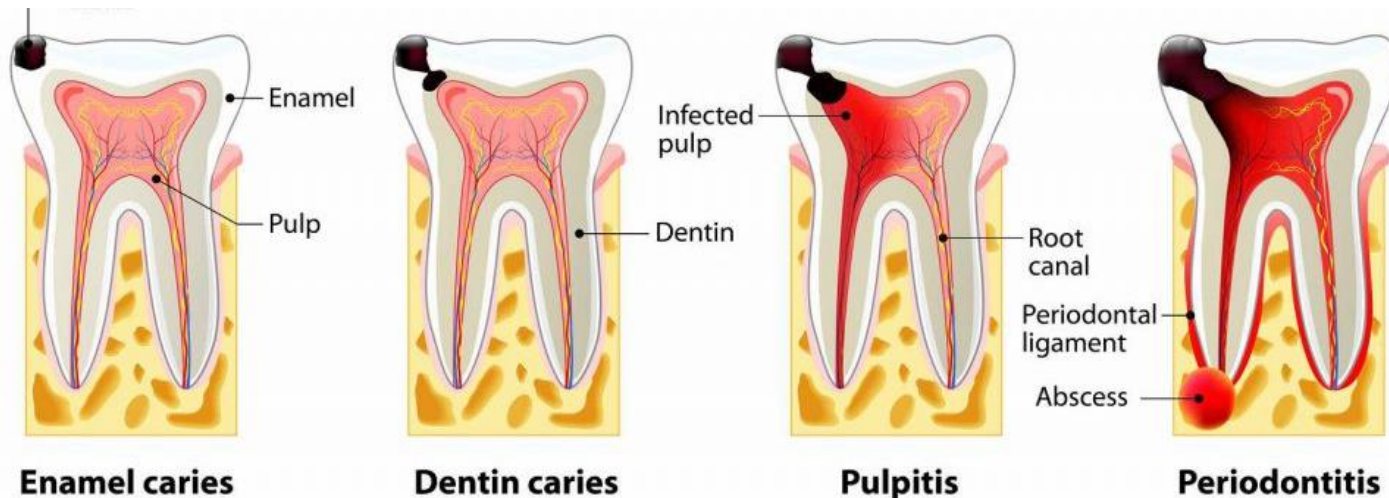
The presence of dental plaque predisposes to the onset of the two main dental pathologies:

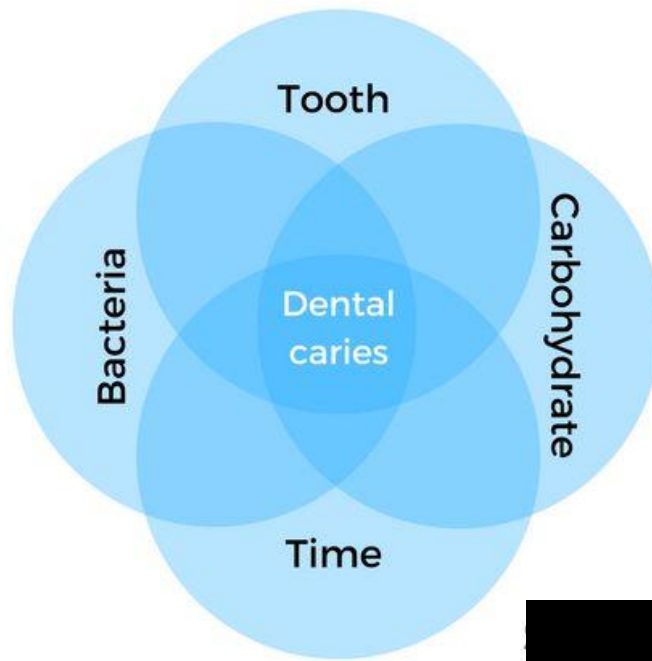
1. Dental caries and
2. Periodontal disease.

DENTAL CARIES

Dental caries is associated with the metabolism of dietary sugar to acid by predominantly Gram-positive organisms (e.g. *Streptococcus mutans*) in supragingival plaque.

Frequent intake of sugar induces a cariogenic plaque in which organisms capable of surviving at low pH are favoured





PERIODONTAL DISEASE



Periodontal disease is associated with the persistent presence of mature plaque at the gingival margin and worsened by an increase in Gram-negative organisms and obligate anaerobes, which are favoured by the environment of periodontal pockets.

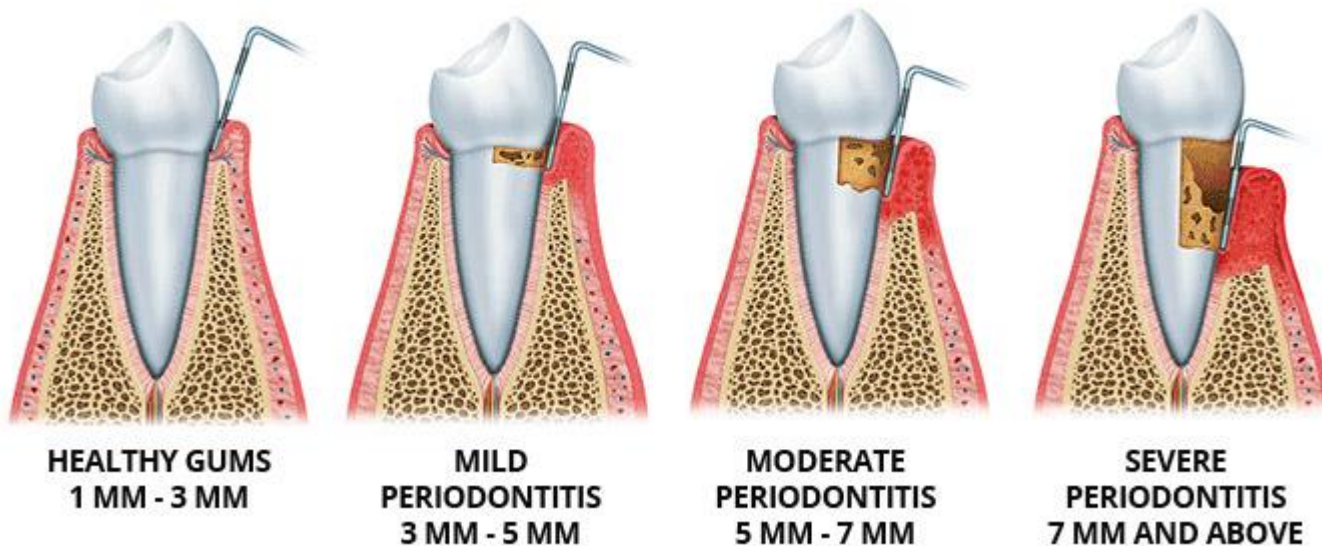
GINGIVAL INFLAMMATION AND PERIODONTAL DESTRUCTION ARE THE RESULT OF DIRECT ACTION OF BACTERIAL PRODUCTS (SUCH AS PROTEASES) AND INDIRECT PROMOTION OF POTENTIALLY DAMAGING IMMUNE RESPONSES.





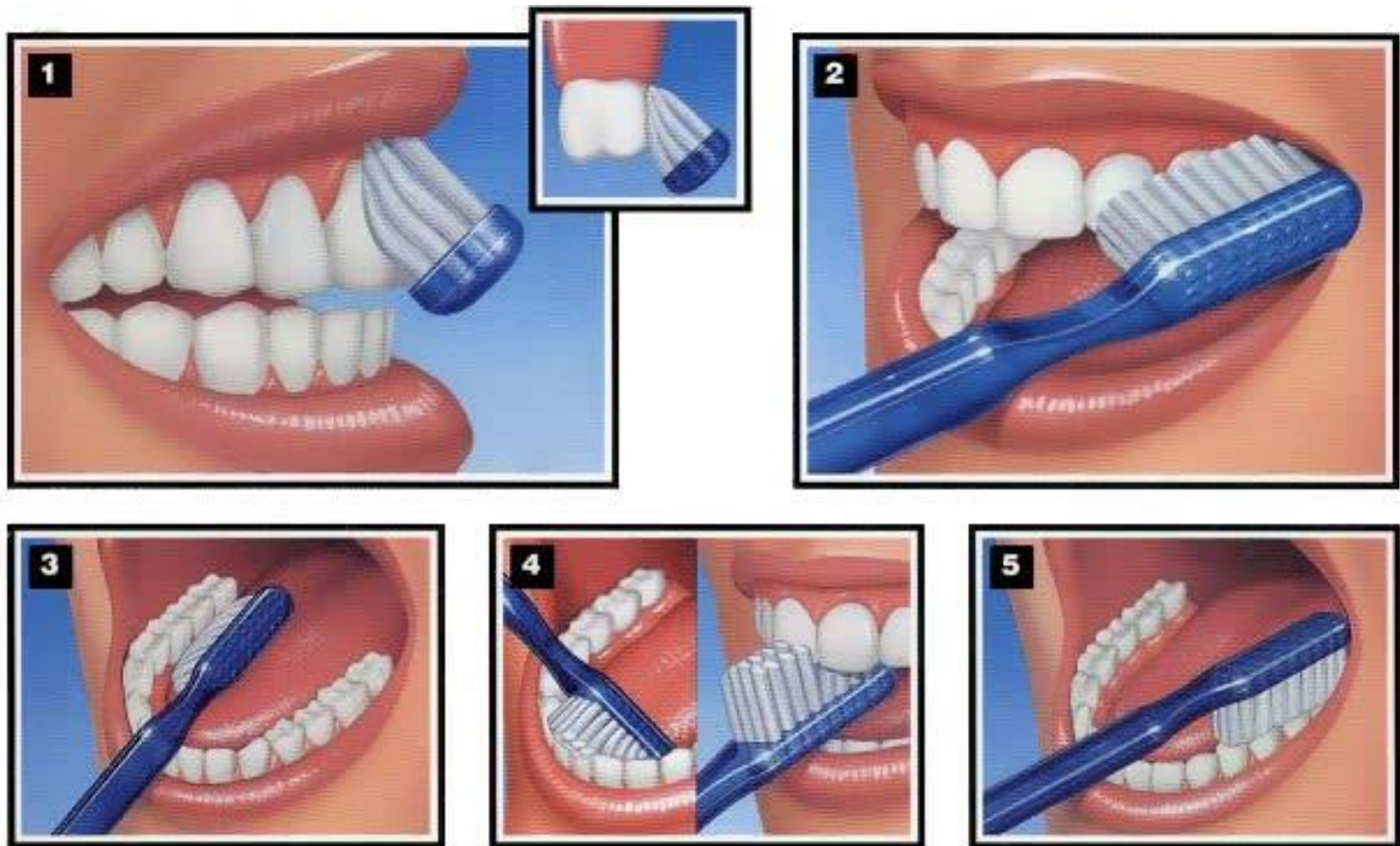
PATHOGENIC POTENTIAL MAY BE MODIFIED BY THE USE OF PERSONAL ORAL HYGIENE PRODUCTS SUCH AS TOOTHPASTES, MOUTH RINSES AND GELS CONTAINING ANTIMICROBIALS, OR DELIVERY OF ANTIBIOTICS AS LOCAL OR SYSTEMIC TREATMENTS BY THE DENTAL PRACTITIONER.

STAGES OF GUM DISEASE AND POCKET DEPTHS

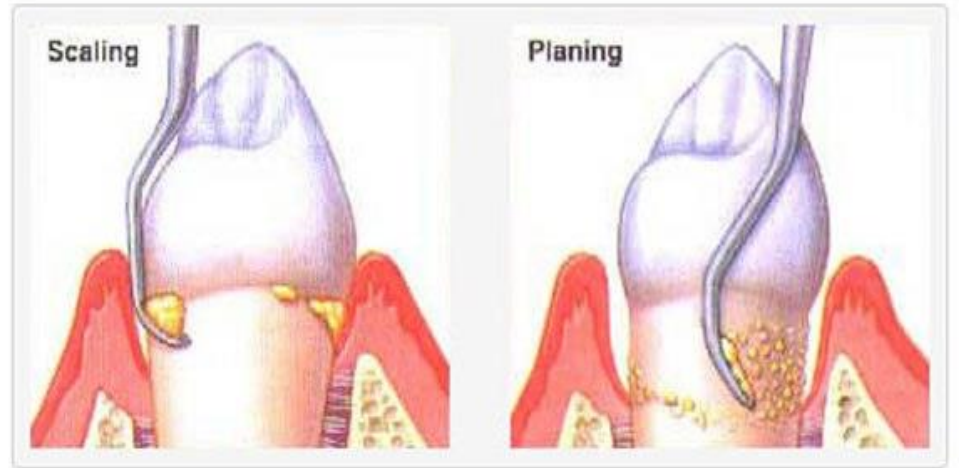


The challenge of dental plaque can be limited by reducing dietary sugar intake, mechanical removal of the plaque and modifying the pathogenic potential.

Toothbrushing, flossing or professional scaling and root planing may achieve mechanical removal









THANK YOU